bomblab □□

□□□□□□

□□□2024201593

□□	phase_1	phase_2	phase_3	phase_4	phase_5	phase_6	secret_phase
6	1	1	1	1	1	1	0

scoreboard □□□

2024201593	0	2	0	0	0	1	0	0	0	0	0	0	0	3	6

□□□□

phase_1

```
If abstraction is the definition of beauty, are those of us chasing after
clarity a representation of ugly?
```

□□□□□□□□call□□strings_not_equal□□□□□□□□□□□□□□□□□□□□□x/s $rsi□□□□□□□□□□□□□□□

phase_2

```
349703  719796  252009  471328
```

□python□□□□□□

```python
def phase_2(input_str):
    #sscanf□□4□□□□□□147d-1489□□
    try:
        inputs = list(map(int, input_str.split()))
    except ValueError:
        explode_bomb()

    #□□□□□□□□□□4:cmp $0x4,%eax □□
    if len(inputs) != 4:
        explode_bomb()

    #matA□matB
    matA = []
    matB = []

    #□□□□□□
```

```python
    result = []
    for row in matA:
        sum_val = 0
        for i in range(3):
            sum_val += row[i] * matB[i][0]
        result.append(sum_val)

    #□□□□□□□
    for input_val, target_val in zip(inputs, result):
        if input_val != target_val:
            explode_bomb()

    print(1)
```

□□□□□□□AB□□

phase_3

```
5 -429
```

□python□□□□□□□

```python
def phase_3(input_str):
    try:
        parts = input_str.strip().split()
        if len(parts) != 3:
            explode_bomb()
        idx = int(parts[0])
        char = parts[1]
        val = int(parts[2])
        # □□□□□□□□□
        if len(char) != 1:
            explode_bomb()
    except (ValueError, IndexError):
        explode_bomb()

    #□□□□□ idx □□□□7
    if idx > 7:
        explode_bomb()

    #□□□
    answer_char = None
    answer_val = None
    mask = 0x20

    if idx == 0:
        answer_char = 0x76
        answer_val = 0xfc
    elif idx == 1:
```

```
            answer_char = 0x76
            answer_val = 0xfc
        else:
            explode_bomb()

        xor_char = ord(char) ^ mask
        if xor_char != answer_char:
            explode_bomb()

        if val != answer_val:
            explode_bomb()

        print(1)
```

□□ p/x $eax □□mask□□cmpl□□□□□□□□□□□□7□□□□□7□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□input_val2□□□□□ 0xfc□252 15b5: je 169c□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

phase_4

```
31 CA
```

□□□□□□□□□□func4_1(5) □□□□□□□□□□□□□□□□□□□□□□□ 2 * □□□□ + 1□func4_2□□□□□□□□□□□□□2□□□□□□□□□□python □□□□□□

```
def func4_1(n):
    if n <= 0:
        return 0
    if n == 1:
        return 1
    return 2 * func4_1(n - 1) + 1

def func4_2(n, target, start, end, aux, buffer):
    if n == 1:
        # n=1 □□□□□ start □ end □□□□
        buffer.append(chr(start))
        buffer.append(chr(end))
        return
    # □□ limit = func4_1(n-1)
    limit = func4_1(n - 1)
    if target <= limit:
        #(start, aux, end)
        func4_2(n - 1, target, start, aux, end, buffer)
    elif target == limit + 1:
        #start\end
        buffer.append(chr(start))
        buffer.append(chr(end))
        return
    else:
        #(aux, end, start)
```

```python
        new_target = target - limit - 1
        func4_2(n - 1, new_target, aux, end, start, buffer)

def phase_4(input_str):
    try:
        parts = input_str.strip().split()
        if len(parts) != 2:
            explode_bomb()
        input_int = int(parts[0])
        input_str = parts[1]
    except (ValueError, IndexError):
        explode_bomb()

    target_int = func4_1(5)
    if input_int != target_int:
        explode_bomb()

    #□□□□□□□□2
    if len(input_str) != 2:
        explode_bomb()

    #
    buffer = []
    # edi=5, esi=0x16(22), edx=0x41('A'), ecx=0x43('C'), r8d=0x42('B'),
r9=buffer
    func4_2(n=5, target=22, start=65, end=67, aux=66, buffer=buffer)
    target_str = ''.join(buffer)

    if input_str != target_str:
        explode_bomb()

    print(1)
```

□□□ABC□□□□□□□□□□+□□□□□□□□□□□□□

phase_5

```
>74598
```

```python
def phase_5(input_str):
    if len(input_str) != 6:
        explode_bomb()

    target_array = []

    target_str =
    result = []
    for char in input_str:
        char_ascii = ord(char)
```

```
        index = (char_ascii + 0xf) & 0xf
        result_char = chr(target_array[index])
        result.append(result_char)

    if ''.join(result) != target_str:
        explode_bomb()
```

□□□□□□□□□6□□□□□□□□□□□□□□□□□□□□ASCII□□□ 0xf □□ 0xf □□□

phase_6

```
6 1 3 2 4 5 defuse
```

□□□

```
(gdb) set $node_head = 0x55555555A210
(gdb) set $cur = $node_head
(gdb) set $n = 1
(gdb) while $n <= 6
 >printf "□□%d | □□=0x%lx | □□(val)=%d\n", $n, $cur, *(int*)$cur
 >set $next = *(long*)($cur + 8)
 >if $next == 0
  >break
  >end
 >set $cur = $next
 >set $n = $n + 1
 >end
□□1 | □□=0x55555555a210 | □□(val)=777
□□2 | □□=0x55555555a220 | □□(val)=73
□□3 | □□=0x55555555a230 | □□(val)=371
□□4 | □□=0x55555555a240 | □□(val)=570
□□5 | □□=0x55555555a250 | □□(val)=531
□□6 | □□=0x55555555a160 | □□(val)=610
```

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□7□□□

secret phase

```

```

□□□□□□□□□□□□□□□□□ □□□□□ 0x55555555a1a0 : 0x0000010000010000 0x000055555555a1b0 0x55555555a1b0 : 0x0100000001000000 0x000055555555a1c0 0x55555555a1c0 : 0x0000010000010001 0x000055555555a1d0 0x55555555a1d0 : 0x0000000000000001 0x000055555555a1e0 0x55555555a1e0 : 0x0001000100000100 0x000055555555a1f0 0x55555555a1f0 : 0x0000000101000001

0x000055555555a200 0x55555555a200 : 0x0100010000000000 0x000055555555a150 0x55555555a150 : 0x0000000000000100 0x0000000000000000 □□□□□□□□ [0,0,1,0,0,1,0,0], # row0 (□0) [0,0,0,1,0,0,0,1], # row1 (□1) [1,0,1,0,0,1,0,0], # row2 (□2) [1,0,0,0,0,0,0,0], # row3 (□3) [0,1,0,0,1,0,1,0], # row4 (□4) [1,0,0,1,1,0,0,0], # row5 (□5) [0,0,0,0,0,1,0,1], # row6 (□6) [0,1,0,0,0,0,0,0], # row7 (□7) □□□□ (gdb) print *(int[8])*($rsp) $5 = {-2, -1, 1, 2, 2, 1, -1, -2} (gdb) print *(int[8])*($rsp+0x20) $6 = {1, 2, 2, 1, -1, -2, -2, -1} (gdb) print *(int[8])*($rsp+0x40) $7 = {-1, 0, 0, 1, 1, 0, 0, -1} (gdb) print *(int[8])*($rsp+0x60) $8 = {0, 1, 1, 0, 0, -1, -1, 0}□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

1. □□□□□□□□3□□□idx
2. □□□□□□□ = (esi+C[idx], edx+D[idx])
3. □□□□□□□□□□□□□□□1□□□□
4. □□□□□□ = (esi+A[idx], edx+B[idx])
5. □□□□□□□□□□□□□□1□□□□ □□□□□□□□□□□□1□□□□□□□□□
6. □□□□□□□□□□□□□□20□□□□□□4□7□

□□□□□□□□python□□

```
from collections import deque

# □□□□□□1=□□0=□□□□
map_grid = [
    [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],  # □0
    [1, 0, 0, 1, 0, 0, 1, 0, 0, 1],  # □1
    [1, 0, 0, 0, 1, 0, 0, 0, 1, 1],  # □2
    [1, 1, 0, 1, 0, 0, 1, 0, 0, 1],  # □3
    [1, 1, 0, 0, 0, 0, 0, 0, 0, 1],  # □4
    [1, 0, 1, 0, 0, 1, 0, 1, 0, 1],  # □5□□□□□5□□8□
    [1, 1, 0, 0, 1, 1, 0, 0, 0, 1],  # □6
    [1, 0, 0, 0, 0, 0, 1, 0, 1, 1],  # □7
    [1, 0, 1, 0, 0, 0, 0, 0, 0, 1],  # □8
    [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]  # □9
]


A = [-2, -1, 1, 2, 2, 1, -1, -2]
B = [1, 2, 2, 1, -1, -2, -2, -1]
C = [-1, 0, 0, 1, 1, 0, 0, -1]
D = [0, 1, 1, 0, 0, -1, -1, 0]

# □□□□□□
DEST_ESI = 5
DEST_EDX = 8

# □□□□□□□□□□<20□
MAX_LENGTH = 19


def find_valid_number_string():
    """
    □□□□□□□□□□□□□□□(1,1)□□□□□□□□□□□(5,8)□□□<20
    """
```

```
    # □□□BFS□□□□(□□esi, □□edx, □□□)
    queue = deque()
    queue.append((1, 1, ""))

    # □□□□□□□□□□□□□□□□□□□
    visited = set()
    visited.add((1, 1))

    while queue:
        current_esi, current_edx, num_str = queue.popleft()

        # □□□□□□□□□□□□□□
        if len(num_str) >= MAX_LENGTH:
            continue

        # □□□□□□□□□□□0-7□□□A/B/C/D□□□□□
        for i in range(8):
            # □□1□□□□□□□□□□□□□□□□
            temp_esi = current_esi + C[i]
            temp_edx = current_edx + D[i]
            # □□□□□□□□□□□□□□□□□□□
            # if not (0 <= temp_esi < 10 and 0 <= temp_edx < 10):
            #     continue  # □□□□□□□□□
            if map_grid[temp_esi][temp_edx] == 1:
                continue  # □□□□□□□□□

            # □□2□□□□□□□□□□□□□□□
            new_esi = current_esi + A[i]
            new_edx = current_edx + B[i]
            # □□□□□□□□□□□□□□□□□□□
            # if not (0 <= new_esi < 10 and 0 <= new_edx < 10):
            #     continue
            if map_grid[new_esi][new_edx] == 1:
                continue

            # □□□□□□□
            new_num_str = num_str + str(i)

            # □□□□□□□□□□
            if new_esi == DEST_ESI and new_edx == DEST_EDX:
                return new_num_str

            # □□□□□□□□□□□□□
            if (new_esi, new_edx) not in visited:
                visited.add((new_esi, new_edx))
                queue.append((new_esi, new_edx, new_num_str))

    # □□□□□□□□□□□□
    return "□□□□□□<20□□□□□□□"


# □□□□□□□□□□
if __name__ == "__main__":
```

```
    result = find_valid_number_string()
    print(f"□□□□□□□□□□{result}")
```

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□□/□□/□□/□□

□6□phase□□□□□□□□8□□□□□□□□□□secret_phase□□□ □□□□□□□□□□□□□□□□□□□□□□1□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□break□□□ □□□□□□□□□□□□□bomblab□□□□□□□□□□□□□□□□□□□□□□□□□□□

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